

Comparative Analysis of Task Scheduling Algorithms in Multi-Robot Material Handling Systems

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EXTENDED ABSTRACT

This study conducts a rigorous comparative performance analysis of three fundamental task scheduling paradigms—First-In, First-Out (FIFO), Priority-Based Scheduling, and Round-Robin—applied to automated multi-robot material handling cells. Using both discrete-event mathematical modeling and high-fidelity physics simulations with mobile bases and 6-DOF manipulators, the authors investigate waiting time distributions, robot utilization rates, and makespan under stochastic arrival loads. Results demonstrate that Priority-Based scheduling achieves a 62% reduction in critical part waiting latency while maintaining total system throughput.

Keywords: Task Scheduling, FIFO, Priority-Based Scheduling, Round-Robin, Multi-Robot Logistics, Resource Utilization

1. Research Objectives & System Model

In modern smart factories, stationary manufacturing fixtures require constant part feeding by autonomous mobile robots. When task arrival rates surge, the choice of scheduling algorithm dictates whether high-value assembly lines stall or remain fully saturated.

The study compares three dispatching policies:

- FIFO (First-In, First-Out): Servicing tasks in strict chronological order of order entry.
- Priority-Based: Organizing tasks into min-heaps indexed by assembly urgency and part perishability.
- Round-Robin: Distributing mobile feeder runs across manufacturing cells in cyclic turns.

2. Mathematical Formulation of Scheduling Metrics

Average Task Waiting Time across N tasks:

$$W_{avg} = (1 / N) \cdot \sum (t_{start, i} - t_{arrival, i})$$

Robot Resource Utilization for robot \$R\$ over total experiment duration \$T_{total}\$:

$$U_R = [(\sum \tau_{busy, R, i}) / T_{total}] \times 100\%$$

Throughput normalized to completed tasks per hour: $TH = (N_{completed} / T_{total}) \times 3600$ (Tasks / Hour)

3. Empirical Benchmark Results

A standardized workload of 50 stochastic parts was evaluated under balanced and bursty arrivals.

Scheduling Method	Makespan (s)	Avg Wait Time (s)	Urgent Task Wait (s)	Manipulator Util (%)
FIFO	842.1	215.4	198.6	72.4%
Priority-Based	848.5	194.2	75.3 (-62.1%)	78.1%
Round-Robin	851.0	228.7	210.2	71.0%

4. Synthesis & Recommendations

- Priority-Based scheduling is strongly superior in industrial environments with heterogeneous part values.
- Round-Robin guarantees fair service distribution across multiple feeding bays but penalizes critical batches.
- FIFO is computationally trivial but prone to severe head-of-line blocking.

DIRECT RELEVANCE & SYNTHESIS FOR TEAM 3 PROJECT

Provides the complete mathematical formulation and baseline performance metrics used in Team 3's Project (Activity) deliverables, specifically validating our comparative analysis between FIFO, Priority, and Round-Robin algorithms.